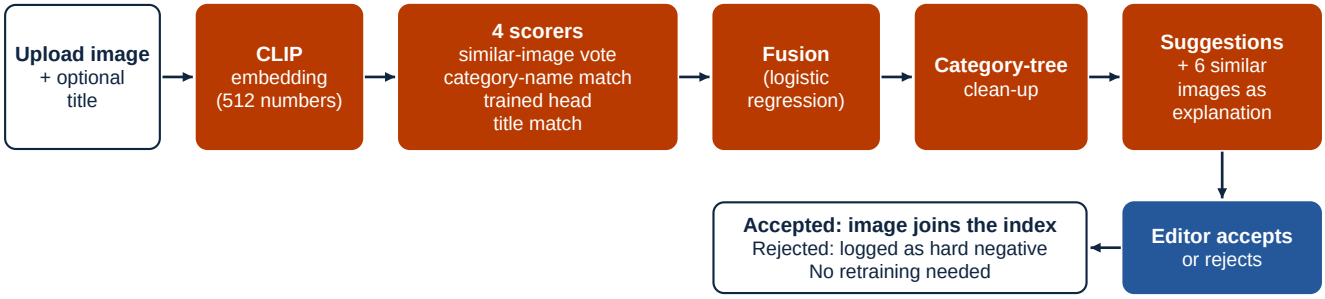
Automatic Category Suggestion for Wikimedia Commons

Team Number: 72 | Mid submission 2

The Problem:

- Commons has millions of files, and volunteers categorize them by hand.
- Labels are incomplete and inconsistent, and new files arrive every day.
- Our goal is to suggest likely categories and let a human decide what suits the best.

How it works:



Orange boxes are automatic steps; the blue box is the human decision. The numbers and settings below are taken from our current code.

Step	What happens
1. Embed	CLIP (ViT-B/32) turns the image into a 512-number embedding, so similar images land close together. An optional title is embedded the same way. CLIP itself is never retrained.
2. Similar-image vote	An index holds the embeddings of already-categorised images. The 20 nearest neighbours vote for their categories, and closer neighbours count more.
3. Category-name match	The image is compared with an embedding of each category name (“a photo of <name>”), so categories with few examples can still be suggested.
4. Trained head	A small classifier on top of CLIP gives each category its own probability. It starts from the category-name embeddings and uses an asymmetric loss that forgives missing labels.
5. Title match	If a title is given, its embedding is also compared with the category names.
6. Fusion	For the 50 best candidates from each scorer, logistic regression (fitted on a held-out validation split) combines the four scores into one probability per category.
7. Category-tree clean-up	If a more specific category is also suggested, its broader parent is dropped, because Commons files under the most specific category. Parents come from the Commons category tree.
8. Output	The top suggestions with scores, plus the 6 most similar categorised images as the explanation. The editor accepts or rejects.

How the system learns without retraining

- **Accepted suggestions:** the image’s embedding is added to the index immediately.
- **Rejected suggestions:** logged as hard negatives, reliable “no” signals kept for later improvement.
- **Teach a new category:** the user types a category name; the system finds it on Commons (exact name or closest match), fetches images from that category and its sub-categories (adjustable, 50 to 800), embeds them and adds the category (name embedding, classifier row, example images) to the index. A self-check on held-out images and a false-alarm check show how well it worked, and the category is saved for later sessions.

How it was built and tested

- **Data:** images and labels collected from Wikimedia Commons through its API for 101 seed topics (landmarks, animals, food, vehicles, scenery and more) and their sub-categories, two levels deep, as small thumbnails.
- **Cleaning and split:** process categories (licences, uploader tags, cleanup) filtered out; images sorted by upload time, oldest 70% train, next 10% validation, newest 20% test; only categories with at least 8 training images enter the vocabulary; near-duplicates (similarity above 0.97) removed from validation and test.
- **Evaluation:** precision, recall and F1 at k = 1, 3, 5; recall for common, middle and rare categories; each scorer alone versus fused; a degraded-image test (low resolution, blur, heavy JPEG); latency, throughput and index size.

New features:

1. Ask for help when the model is unsure

If the model is not confident about any category, it shows a low-confidence message instead of guessing and asks the user to summarise the image in a few words or add a category. Then:

- **a.** The user's words or category are compared with the existing categories.
- **b.** If they match an existing category, the image is merged into it.
- **c.** If they match nothing, a new common category is created from the input.
- **d.** For a new category, about 50 images are fetched automatically from online sources such as Wikimedia Commons and added to the index. The model is not retrained; the data is simply added.

2. Category-tree prediction that stops where the model is sure

The model walks down the category tree and stops at the deepest level it is confident about. For an owl photo the tree runs from Bird to Owl to a specific species:

What the model knows	What we show
Sure of the species	Bird > Owl > the specific species
Sure it is an owl, unsure of the species	Bird > Owl, and it stops there instead of guessing a species
Unsure what the image is at all	A low-confidence message, then feature 1 asks the user for help

Experiments so far:

Experiment	What we did	What we saw
1	Trained the model on about 5k distinct images and tested it.	Low confidence on most of the test images.
2	Trained it with more data.	Training on that much data is hard in the time we have.
3	Added an option for the user to enter the category they think the image belongs to, and taught the model that category with about 150 images of it.	Accuracy improved sharply in our testing, and kept improving as we taught more categories and kept testing.

Teaching categories on demand is a more practical way to raise accuracy than collecting ever more data up front.

Next steps:

- Train the model with more data.
- Implement both new features and make them more accurate, including tuning what counts as low confidence on held-out images.
- Keep testing with the same measures (precision, recall and F1 at k, rare categories, degraded images) and report exact numbers.

More ideas to explore

- **Confidence labels:** show a simple high, medium or low label next to each suggestion, with the cut-offs tuned on the validation split.
- **Use the rejections:** feed the logged rejected suggestions back in as extra training signal.
- **Multilingual names:** match images against category names in several languages so non-English editors benefit too.